Supplementary material

Does starting age predict who stays? Competition debut and long-term participation in youth track and field

Anonymised manuscript (Brief Report). Abbreviations: CI, confidence interval; COVID-19, coronavirus disease 2019; IQR, interquartile range; OR, odds ratio; Q1 and Q4, first and fourth quarter of the birth year; SE, standard error.

Table S1. Sample flow.

Step	n
Athletes born 2001–2008 in the registry extract	33 666
Result rows extracted (2011 – 29 Oct 2026)	675 707
excluded: non-approved rows	1 526
excluded: rows below age 10	116
Debutants with an approved result at age $\geq$10	28 339
excluded: missing sex	865
set aside: debut at age $\geq$17 (sensitivity analyses)	731
Debutants at ages 10–16 with known sex	26 743
primary sample: born 2001–2005	17 496
descriptive only: born 2006–2008	9 247
Active seniors in the primary sample	546

Table S2. Senior conversion by debut age and sex (born 2001–2005), % (Wilson 95 % CI).

Debut age	Girls, n	Girls, % (CI)	Boys, n	Boys, % (CI)
10	3 485	3.4 (2.8–4.0)	3 141	4.3 (3.6–5.1)
11	1 860	2.2 (1.6–2.9)	1 756	2.7 (2.1–3.6)
12	1 263	1.8 (1.2–2.7)	1 266	2.3 (1.6–3.3)
13	879	2.0 (1.3–3.2)	812	3.8 (2.7–5.4)
14	789	2.0 (1.3–3.3)	958	2.7 (1.9–3.9)
15	326	2.5 (1.2–4.8)	537	4.8 (3.3–7.0)
16	158	5.7 (3.0–10.5)	266	7.1 (4.6–10.9)

Sex $\times$ debut-age interaction (linear): OR 0.92 per year for girls relative to boys, 95 % CI 0.82–1.02. Full coefficient tables for the primary models are available with the analysis code.

Table S3. Retention k years after the debut season: proportion (Wilson 95 % CI). Denominators equal the debut-age n except debut 16 at k = 5 (n = 389).

Debut age	n	k=1	k=2	k=3	k=4	k=5
10	6 626	.512 (.500–.524)	.379 (.367–.390)	.288 (.277–.299)	.218 (.208–.228)	.169 (.160–.178)
11	3 616	.398 (.382–.414)	.269 (.254–.283)	.188 (.175–.201)	.131 (.121–.143)	.097 (.087–.107)
12	2 529	.372 (.353–.391)	.234 (.218–.251)	.143 (.130–.157)	.094 (.083–.106)	.071 (.061–.081)
13	1 691	.378 (.356–.402)	.226 (.207–.247)	.150 (.134–.168)	.100 (.087–.115)	.068 (.057–.081)
14	1 747	.340 (.318–.363)	.128 (.113–.144)	.080 (.068–.093)	.055 (.045–.067)	.033 (.025–.042)
15	863	.256 (.228–.286)	.125 (.105–.149)	.080 (.064–.100)	.049 (.036–.065)	.044 (.032–.060)
16	424	.349 (.305–.396)	.210 (.174–.251)	.111 (.084–.144)	.066 (.046–.094)	.059 (.040–.087)

Table S4. Competing at equal career length: proportion with at least two competition days in a single season at least 4, 5 and 6 years after the debut season, by debut age (n = athletes for whom that season was fully observable by 2025), with the adjusted OR per year of later debut (sex, birth cohort, club size).

Debut age	4 years: % (n)	5 years: % (n)	6 years: % (n)
10	20.4 (6 626)	15.7 (6 626)	12.0 (6 626)
11	12.4 (3 616)	8.8 (3 616)	6.4 (3 616)
12	8.5 (2 529)	6.2 (2 529)	4.0 (2 529)
13	8.7 (1 691)	6.2 (1 691)	4.1 (1 691)
14	4.5 (1 747)	2.9 (1 747)	2.2 (1 747)
15	4.9 (863)	3.8 (863)	2.6 (779)
16	6.6 (424)	5.1 (389)	3.2 (317)
Linear trend, OR per +1 year	0.70 (0.68–0.72)	0.69 (0.66–0.72)	0.66 (0.63–0.69)

Table S5. Robustness analyses. Estimates are for debut age (per +1 year, linear) unless stated; outcome is senior activity unless stated.

Analysis	n	Events	OR (95 % CI)
Primary linear trend	17 496	546	0.99 (0.94–1.04)
Excluding debuts in 2020–2021	17 305	530	0.95 (0.90–1.01)
Debut at 16 vs. debut at 10 (categorical), primary	17 496	546	1.73 (1.15–2.59)
excluding debuts in 2020–2021	17 305	530	1.32 (0.79–2.19)
adding the COVID-19 debut flag	17 496	546	1.38 (0.86–2.19)
cohorts born 2001–2003	10 667	349	1.35 (0.80–2.26)
cohorts born 2004–2005	6 829	197	3.08 (1.59–5.96)
Excluding the 2001 cohort (debut at 10 in 2011)	14 077	434	0.99 (0.94–1.05)
Including athletes without registered sex	18 045	547	0.99 (0.94–1.04)
Outcome: $\geq$1 result at age $\geq$20	17 496	694	1.00 (0.96–1.05)
Outcome: any result at age $\geq$17 (debuts 10–14)	16 209	1 621	0.97 (0.93–1.00)
Adding debut-season competition days	17 496	546	0.94 (0.89–1.00)
Cohorts born 2001–2003	10 667	349	0.98 (0.92–1.05)
Cohorts born 2004–2005	6 829	197	0.99 (0.91–1.08)
Debut at 17–19 vs. debut at 10 (categorical)	17 898	595	3.36 (2.42–4.67)
Adding the COVID-19 debut flag (linear trend)	17 496	546	0.95 (0.90–1.01)
Club-clustered robust SE (trend)	17 496	546	0.99 (0.92–1.06)

Table S6. Characteristics of the primary sample (born 2001–2005, debut at ages 10–16), by debut group.

Characteristic	Debut 10–11	Debut 12–13	Debut 14–16
n	10 242	4 220	3 034
Female, %	52.2	50.8	42.0
Exact birth date available, %	38.6	63.8	88.5
Born Q1, % of those with birth date	29.2	26.1	27.1
Born Q4, % of those with birth date	19.9	21.9	19.8
Competition days in debut season, median (IQR)	1 (1–3)	1 (1–3)	1 (1–2)
Career length, seasons, median (IQR)	2 (1–4)	1 (1–3)	1 (1–2)
Debut season 2020–2021, %	0.0	0.0	6.3

By construction, cohorts born 2001–2005 could only experience a 2020–2021 debut at ages 15 and 16.

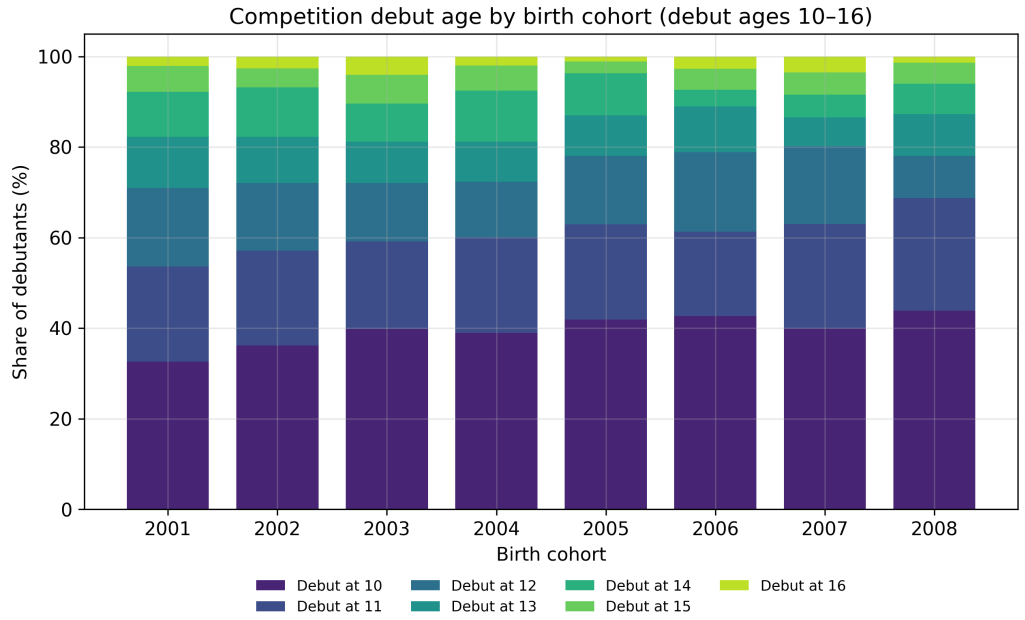


Figure S1. Competition debut age by birth cohort (all debutants born 2001–2008, debut ages 10–16).

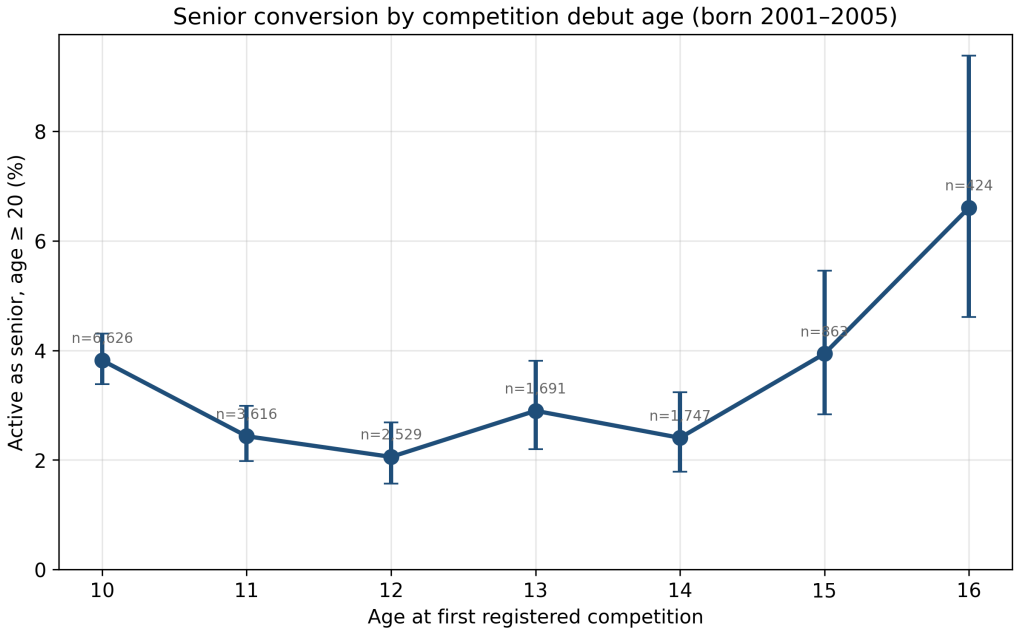


Figure S2. Senior conversion by competition debut age, with Wilson 95 % confidence intervals (born 2001–2005).

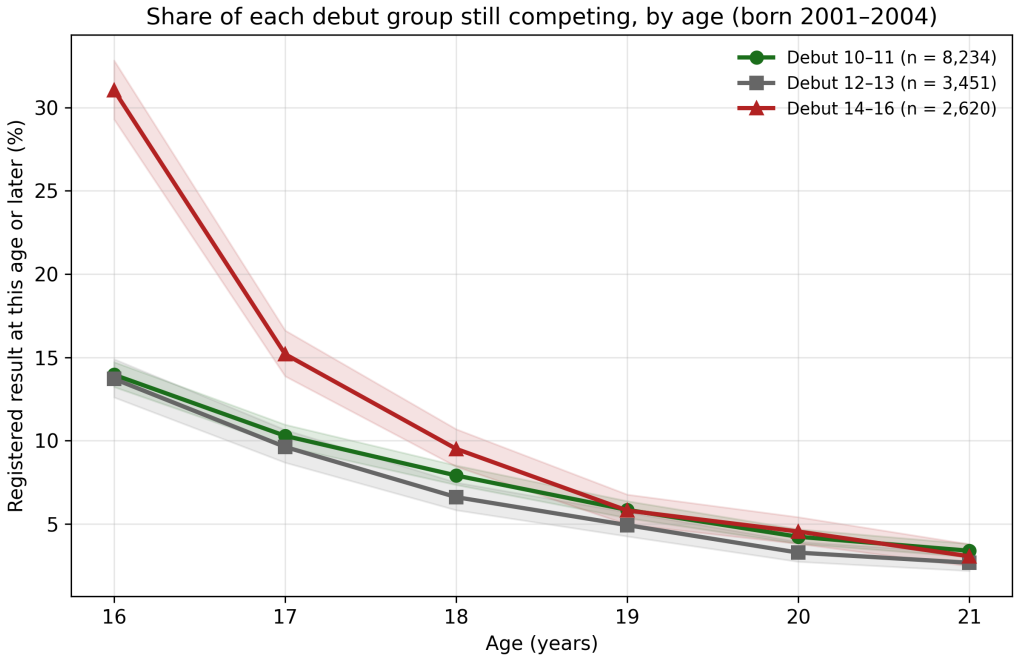


Figure S3. Share of each debut group with a registered result at that age or later, cohorts born 2001–2004 (followed to age 21 for everyone), with Wilson 95 % confidence bands. At 16 the late group includes the athletes who debuted at that age.